

Roll No

CE-601 (GS)**B.E. VI Semester** Examination, June 2020**Grading System (GS)****Theory of Structures-II****Time : Three Hours****Maximum Marks : 70**

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Determine the end moments of the member of the frame as shown in Figure-1 EI is constant.
(Use Moment Distribution Method)

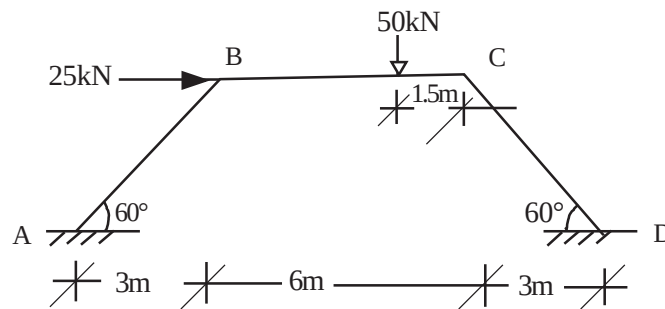


Figure-1

2. Analyse the continuous beam shown in Figure 2 by flexibility matrix method.

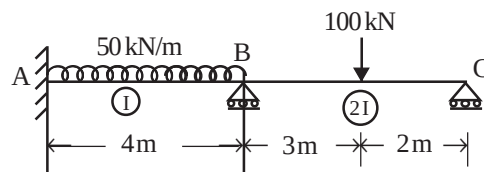


Figure 2

3. Determine the shape factor for triangular section.
4. Enlist various approximate methods used for the analysis of multistoried frames for horizontal and vertical loads stating their limitations.

OR

Explain the portal method, cantilever method and factored method for analysis a building frame subjected to horizontal forces.

[2]

5. Explain the following term (any four):
 - a) Influence line
 - b) Rotation factor
 - c) Muller Breslau principal
 - d) Plastic hinge
 - e) Shape factor
 - f) Distribution factor
6. A two-span continuous beam ABC each of span l is fixed at end A and simply supported at the other end C. Find the collapse load if it is subjected to *u.d.l.* of w /unit length. Take it that the beam is uniform and has plastic moment M_P .
7. How the influence lines can be drawn qualitatively for determinate and indeterminate structures? Explain.
8. Answer any two of the following:
 - a) Write a short note on box culvert method of analysis.
 - b) Differentiate plastic analysis of structure with elastic analysis.
 - c) Write a short note on structural behaviour of tall buildings subjected to lateral loads.
 - d) Write any three differences between force method and displacement method.
